

UME309: Engineering Fluid Mechanics

L	T	P	Cr
2	1	2*	3.0

* Lab every alternate week

Course Objective: Students will be exposed to the basic fundamentals of fluid statics and dynamics, such as mass, momentum and energy conservation and applications of the same.

Syllabus

Fluid statics: Fluid properties and their measurement, Newton's law of viscosity, Newtonian and non-Newtonian fluids, application of hydrostatic pressure, stability of submerged bodies.

Fluid flow fundamentals: Laminar and turbulent flow, Reynolds number, continuity equation, Euler and Bernoulli equation, fluid flow measurement, Navier-Stokes equation, stream and potential functions.

Internal flow: Flow development, pressure drop in laminar and turbulent flow, Darcy equation, application of Moody's diagram, major and minor losses, series and parallel flow.

Dimensionless analysis: The concept of similarity analysis, application of Buckingham Pi theorem, formulation of key dimensionless numbers.

Boundary layer: Boundary layer profile, displacement, momentum and energy thickness, laminar, turbulent and separated flows over flat surfaces.

External flow: Drag and lift, friction and pressure drag, drag coefficients of common geometries.

Laboratory Work

1. Determination of viscosity of water using a capillary tube viscometer.
2. Determination of total pressure of a vertically submerged surface in water.
3. Illustration of laminar, transition and turbulent flow in pipe w.r.t Reynolds number.
4. Verification of the Bernoulli equation.
5. Venturimeter calibration and variation of coefficient of discharge w.r.t. Reynolds number.
6. Determination of head loss due to friction in smooth and rough pipe.
7. Determination of boundary layer, displacement and momentum thickness in wind tunnel.

Course Learning Objectives (CLO)

The students will be able to:

1. Calculate hydrostatic pressure in fluid systems.

2. Formulate and apply mass, momentum and energy equations in fluid flow.
3. Estimate pressure losses drop in different fluid flow regimes.
4. Develop dimensionless numbers to describe fluid interactions.
5. Determine forces in bodies submerged in fluids.

Text Books

1. Cengel, Y. A. and CIMBALA, J.M., Fluid Mechanics, Tata McGraw Hill Publishing Company Limited, New Delhi (2006).
2. White, F. M., Viscous Fluid Flow, McGraw Hill, New York (2006).

Reference Books

1. Bird, R. Byron, Warren E. Stewart, and Edwin N. Lightfoot. Transport Phenomena. New York, NY: John Wiley & Sons , 2nd ed. 2001.
2. Douglas J. F., Gasiorek, J. M. and J. A. Swaffield, Fluid Mechanics, Addison-Wesley Longman Inc., Edinburgh, U.K (1995).
3. Panton, R.L., Incompressible Fluid Flow, John Wiley & Sons, New Jersey (2005).

Evaluation Scheme

Evaluation component	Weightage
MST	30
EST	40
Sessional	30